Arts et Métiers Institute of Technology
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Glider Wing

PJT

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I Introduction

The mechanical sizing of glider wings is a crucial step in aeronautical design. It ensures the safety, performance, and efficiency of the glider while respecting geometric, mechanical, mass, and financial constraints. This report aims to detail the different steps of the mechanical sizing of a glider wing, focusing particularly on the choice of the wing's thickness and material.

II Objectives

The objectives of this report are as follows:

1. **Analyze geometric and mechanical constraints**: Understand the limitations imposed by the wingspan, ground deflection, maximum stress, and glider mass.
2. **Determine the optimal wing thickness**: Use beam models and equilibrium equations to calculate the thickness e that meets the requirements of the specifications.
3. **Evaluate mechanical performance**: Calculate the deflection, maximum stress, and wing mass as a function of the thickness e.
4. **Choose the appropriate material**: Compare different materials based on their cost, density, and mechanical strength.
5. **Size the spar by evaluating mechanical performance**: We assume it depends only on two parameters:
   (a) Hollow circular (radius R and thickness e)
   (b) Rectangular (height b and width a)

III Determination of the thickness e

1 Beam model

To model the glider wing, we use the Euler-Bernoulli beam model. This model relies on the following assumptions:

- One of the dimensions (the length L) is much larger than the other two (the thickness e and the height h).
- Cross sections are assumed to be rigid and remain perpendicular to the neutral axis.
- The small perturbation assumptions are applied, which allows linearization of the equations.

2 Equilibrium equations

We work on a single wing by symmetry. The global equilibrium equations are:

- Vertical force equilibrium:

$$Y_O + \int q\,dx = 0 \Rightarrow Y_0 = -qL$$

- Moment equilibrium:

$$M_0 + \int xq\,dx = 0 \Rightarrow M_0 = -\frac{qL^2}{2}$$

3 Internal forces

Internal forces are calculated as follows:

- Shear force $T(x)$:

$$T(x) = \int_x^L q\,dx = q(L-x)$$

- Bending moment $M(x)$:

$$M(x) = \int_x^L (\xi - x)q\,d\xi = \frac{q}{2}(x-L)^2$$

4 Equilibrium equation

The equilibrium equation for the deflection $v(x)$ is:

$$EIv'' = M(x) = \frac{q}{2}(x-L)^2$$

With boundary conditions $v(0) = v'(0) = 0$, we get:

$$v(x) = \frac{qx^2}{24EI}\left(x^2 - 4Lx + 6L^2\right)$$

The maximum deflection $v_{\max}$ is given by:

$$v_{\max} = v(L) = \frac{qL^4}{8EI}$$

5 Calculation of the thickness e

To determine the thickness e, the following criteria must be satisfied:

- **Maximum deflection**:
$$v(L) = \frac{qL^4}{8EI} < 30\,\text{cm}$$
- **Maximum stress**:
$$\sigma_{\text{max}} = \frac{qL^2}{2I} y_{\text{max}} < \sigma_e$$
- **Wing mass**:
$$M_a = \rho S L < 150\,\text{kg}$$

6 Second moment of area

For a hollow elliptical section, the second moment of area I is:

$$I_{\text{hollow}} \approx \frac{\pi}{4}\left(3eab^2 + eb^3\right)$$

For a solid elliptical beam, the second moment of area is:

$$I_{\text{solid}} = \frac{\pi}{4} ab^3$$

7 Calculation of stresses on the ground and in flight

Calculations distinguish between hollow and solid ellipses for both section and moment computations:

$$S_{\text{solid ellipse}} = \pi ab, \quad S_{\text{hollow ellipse}} = \pi ab - \pi(a-e)(b-e) \approx \pi(a+b)e$$

Moreover, the wing is subjected to different linear loads on the ground or in flight:
On the ground: $q = -\rho g S$
In flight: $q = p - \rho g S = \frac{M_n g}{2L}$

8 Asymptotic analysis

For small values of e, we can perform an asymptotic analysis:

- **Stress**:
$$\sigma_{\text{max}} \to 2\rho g \frac{(a+b)L^2}{b(3a+b)}, \quad \text{as } e \to 0$$
- **Mass**:
$$M_a \approx \rho L \pi (a+b) e = \alpha e$$
- **Deflection**:
$$v(L) \to \frac{4\rho g (a+b) L^4}{8Eb^2(3a+b)}, \quad \text{as } e \to 0$$

In flight, stress and deflection behave like $\frac{K}{e}$ as e tends to zero.

9 Python program for thickness evaluation

```python
import numpy as np
import matplotlib.pyplot as plt
from math import pi

# Parameters
a, b, L, g = 0.625, 0.2, 11.0, 9.81
M_tot, M_n, v_max_thresh, mass_max = 500, 200, 0.3, 150
e_list = np.logspace(-6, -2, 500)

# Material: Aluminium 7075-T6
E_alu, rho_alu, sigma_e_alu = 71.7e9, 2810, 505e6
mass_alu, sigma_max_alu, deflection_alu = [], [], []

for e in e_list:
    S = pi*(a+b)*e
    I = (pi/4)*b**2*e*(3*a+b)
    M_a = rho_alu*S*L
    q_ground = -rho_alu*S*g
    p = M_tot*g/(2*L)
    q_flight = p - rho_alu*S*g

    sigma = max(abs((q_ground*L**2/(2*I))*b),
                abs((q_flight*L**2/(2*I))*b))
    deflection = max(abs(q_ground*L**4/(8*E_alu*I)),
                     abs(q_flight*L**4/(8*E_alu*I)))

    mass_alu.append(M_a)
    sigma_max_alu.append(sigma)
    deflection_alu.append(deflection)

# Plots
plt.figure(figsize=(10,12))
plt.subplot(3,1,1)
plt.semilogx(e_list,np.array(sigma_max_alu)/1e6,label="Aluminium 7075-T6")
plt.axhline(900,color='r',linestyle='--',label="Max threshold = 900 MPa")
plt.ylabel("Max stress (MPa)")
plt.grid(True); plt.legend(); plt.title("Maximum stress vs Thickness")

plt.subplot(3,1,2)
plt.semilogx(e_list,np.array(deflection_alu)*100,label="Aluminium 7075-T6")
plt.axhline(v_max_thresh*100,color='r',linestyle='--',label="Max deflection = 30 cm")
plt.ylabel("Deflection (cm)")
plt.grid(True); plt.legend(); plt.title("Maximum deflection vs Thickness")

plt.subplot(3,1,3)
plt.semilogx(e_list,mass_alu,label="Aluminium 7075-T6")
plt.axhline(mass_max,color='r',linestyle='--',label="Max mass = 150 kg")
plt.ylabel("Wing mass (kg)"); plt.xlabel("Thickness e (m)")
plt.grid(True); plt.legend(); plt.title("Wing mass vs Thickness")

plt.tight_layout()
```

```
plt.show()
```

Listing 1: Python code to evaluate the thickness

10 Comparison of candidate materials

Material	Equivalent mass	Estimated cost
Aluminium 7075-T6	100.0 kg	1500 €
Steel 25CD4	95.4 kg	382 €
Carbon/epoxy composite	29.3 kg	2930 €

Table 1: Comparison of materials in terms of mass and cost

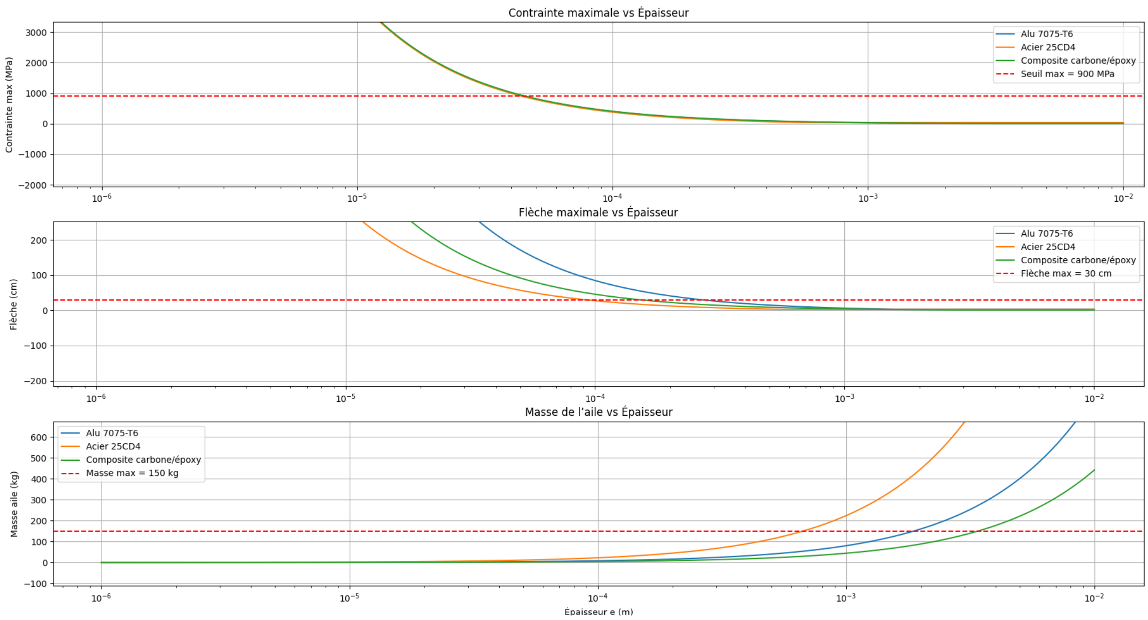


Figure III.1: Evolution of design criteria as a function of thickness

We chose a wing thickness of $e = 0.7\,\text{mm}$ and Aluminium 7075-T6 to design our glider wing.

IV Finite Element Modeling

In this section, we will compare 3 models: an elliptical section beam model, a surface model of an extruded ellipse, and a surface model of the real profile of our glider wing.

Specifications:

- $M = 500\,\text{kg}$: glider mass
- $V = 80\,\text{km/h}$: glider flight speed
- $L = 22\,\text{m}$: total wingspan of the glider
- $c = 1{,}25\,\text{m}$: chord

1 Model 1: Elliptical Section Beam

1.1 Visual Results from Abaqus Simulations

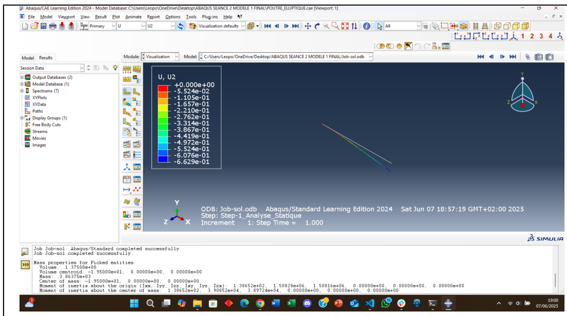
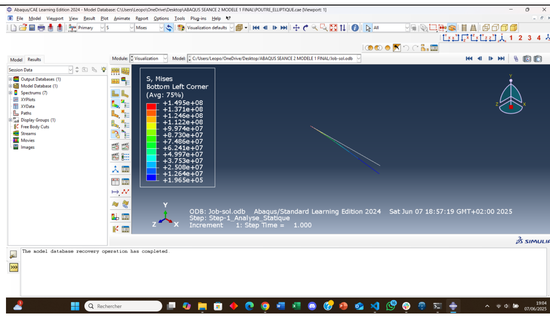
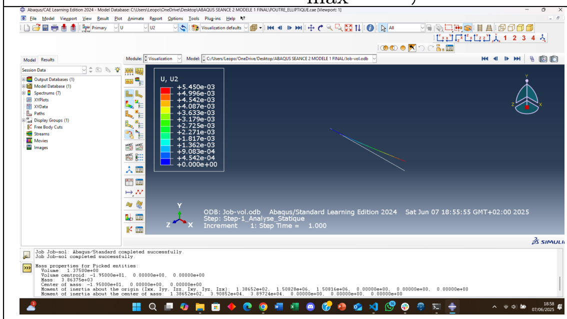
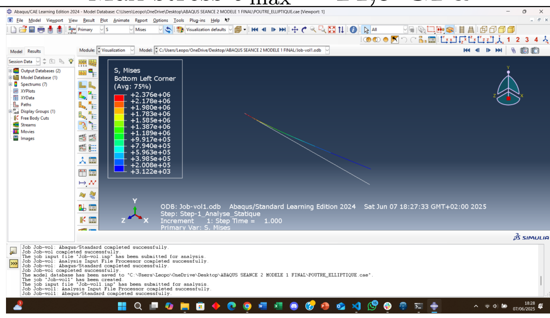
 **Ground simulation – Displacement** Max deflection $v_{\max} =$ **6,2 cm**	 **Ground simulation – von Mises stress** Max stress $\sigma_{\max} =$ **14,9 GPa**
 **Flight simulation – Displacement** Max deflection $v_{\max} =$ **5,45 mm**	 **Flight simulation – von Mises stress** Max stress $\sigma_{\max} =$ **2,376 GPa**

Table 2: Illustrations of Abaqus simulations with maximum deflection and stress

Quantity	Theoretical	Abaqus	Meets specifications
Ground			
Max deflection v_{max}	7 cm	6,2 cm	Yes
Max stress σ_{max}	17 GPa	14,9 GPa	Yes
Mass	12 T	386,375 kg	No (significant difference)
Flight			
Max deflection v_{max}	0,6 mm	5,45 mm	No (deflection too high)
Max stress σ_{max}	0,55 GPa	2,376 GPa	No (stress too high)
Mass	12 T	386,375 kg	No (significant difference)

Table 3: Summary of results for model 1

2 Model 2: Extruded Surface Ellipse

2.1 Visual Results from Abaqus Simulations

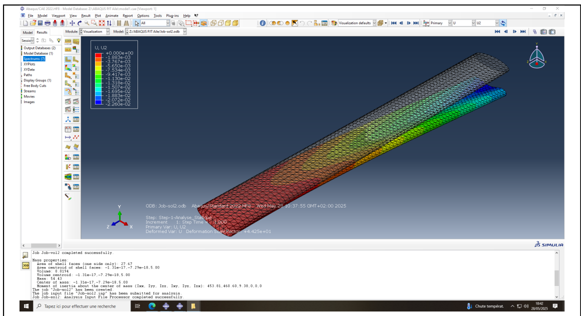
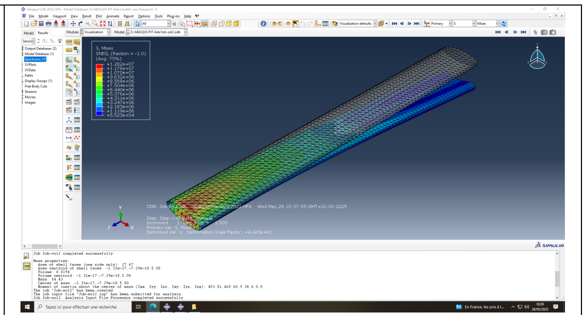
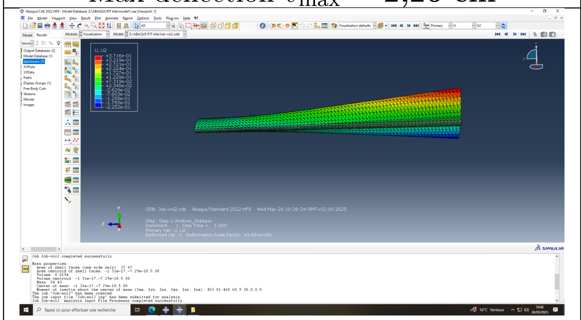
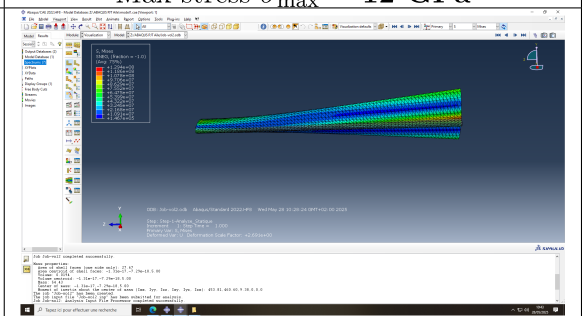
 **Ground simulation – Displacement** Max deflection v_{max} = **2,26 cm**	 **Ground simulation – von Mises stress** Max stress σ_{max} = **12 GPa**
 **Flight simulation – Displacement** Max deflection v_{max} = **3,7cm**	 **Flight simulation – von Mises stress** Max stress σ_{max} = **129 GPa**

Table 4: Illustrations of Abaqus simulations with maximum deflection and stress

Quantity	Theoretical	Abaqus	Meets specifications
Ground			
Max deflection $v_{\max}$	2,7 cm	2,26 cm	Yes
Max stress $\sigma_{\max}$	13,3 GPa	12 GPa	Yes
Mass	54,43 kg	54,43 kg	Yes
Flight			
Max deflection $v_{\max}$	7,08 cm	3,7 cm	Yes
Max stress $\sigma_{\max}$	2,1 GPa	129 GPa	No (overstress)
Mass	54,43 kg	54,43 kg	Yes

Table 5: Summary of results for model 2

The two presented models aim to simulate the structural behavior of a glider wing under different conditions (ground and flight) while respecting the constraints imposed by the specifications.

Model 1, based on an elliptical section beam, shows good agreement between theoretical results and those from the Abaqus numerical model for static ground cases, with maximum deflection and stress below the allowable limits. However, its estimated mass (12 tons) is unrealistic and far exceeds the expected real mass (approximately 386 kg according to Abaqus), invalidating its relevance. In flight, the model also becomes non-compliant: stresses and deflection are too high, making this model unsuitable for accurately representing reality.

Model 2, based on an extruded surface ellipse, shows better consistency between theoretical and numerical aspects. It generally meets the specifications in terms of mass and deformation, both in flight and on the ground. However, a significant overstress is observed in flight in the Abaqus model (129 GPa versus 2,1 GPa theoretical), suggesting that the real distribution of forces in the wing is more complex than analytically modeled.

In conclusion, model 2 is overall more realistic and faithful to the expected physical conditions. Despite the overstress in flight, it remains more relevant than model 1 for representing the behavior of a glider wing as a whole.

3 Model 3: Real Wing Surface Profile

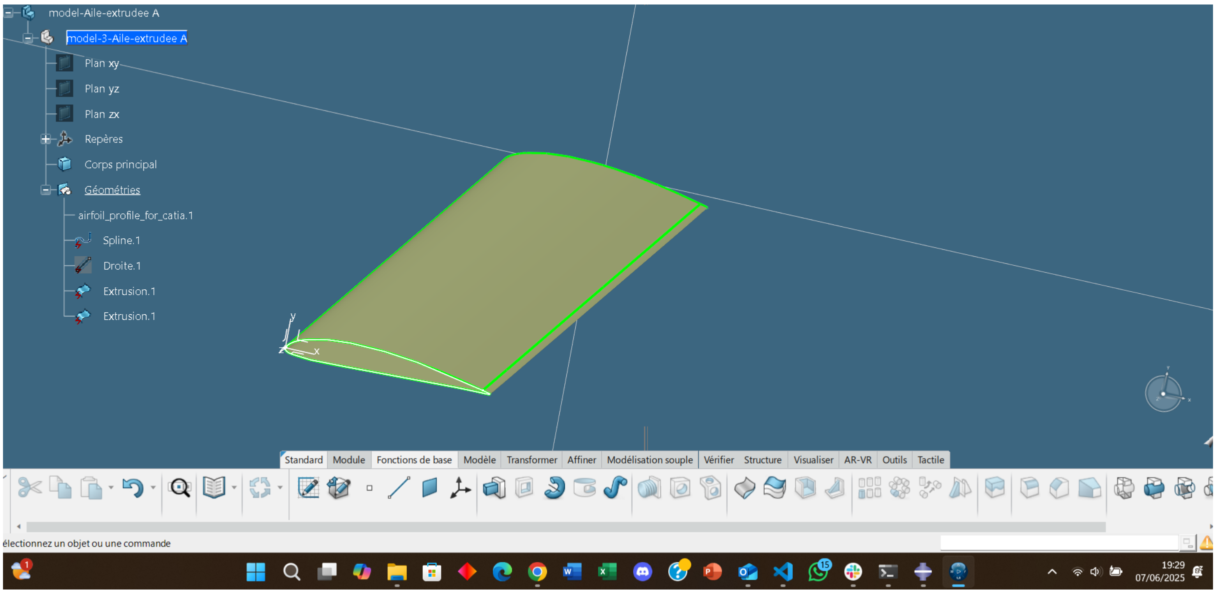


Figure IV.1: Surface model of the wing profile

3.1 Limitations of modeling: absence of real profile simulation

In this study, we were unable to perform an Abaqus simulation of the glider wing's real profile. This limitation is mainly due to the geometric complexity of the profile, as well as the need for a finer and more precise mesh, which requires significant computational resources and careful numerical preparation that we could not implement.

Such a simulation would have allowed direct comparison of theory and numerical results on the real geometry, providing more accurate validation of the wing's mechanical behavior. For example, it could have shown a more realistic stress distribution in the profile thickness, especially at transition zones between the upper and lower surfaces. Theoretically, expected results would have been intermediate between the simplified models (elliptical beam or extruded ellipse) and reality, with better localized deflection and stress, and a more homogeneous force distribution.

This modeling would also have allowed more precise evaluation of critical areas of the wing, such as attachment points or internal reinforcements. Without this simulation, the analysis remains limited to idealized models, capturing only part of the real mechanical phenomena.

4 Comparison of Established Models

Model	Condition	Quantity	Theoretical	Abaqus	Difference
Model 1	Ground	Max deflection $v_{\max}$	7 cm	6,2cm	-
		Max stress $\sigma_{\max}$	17 GPa	$14,9$ GPa	–
		Mass	12 T	386,375 kg	–
	Flight	Max deflection $v_{\max}$	0,6 mm	5,45mm	–
		Max stress $\sigma_{\max}$	0,55 GPa	2,376 GPa	–
		Mass	12T	386,375 kg	–
Model 2	Ground	Max deflection $v_{\max}$	2,7 cm	2,26 cm	-
		Max stress $\sigma_{\max}$	13,3 GPa	12 GPa	–
		Mass	54,43 kg	54,43 kg	-
	Flight	Max deflection $v_{\max}$	7,08 cm	3,7 cm	–
		Max stress $\sigma_{\max}$	2,1 GPa	129 GPa	–
		Mass	54,43 kg	54,43 kg	-

Table 6: Comparison of theoretical and numerical results (Abaqus) under ground and flight conditions

5 Proposed Technological Solution

To address this non-compliance, one solution is to reinforce the section structure by integrating internal reinforcements (e.g., additional spars) that limit local deformation of the section and maintain its geometric shape.

Alternatively, a more advanced modeling approach can be adopted (e.g., Timoshenko beam theory or 3D finite element modeling with bending-torsion coupling) that accounts for real section deformations, optimizing the design without neglecting complex effects.

V Sizing of the Spar

1 Parametric Study of Spar Cross-Sections

We assume that the spar has a cross-section characterized by only two parameters, depending on the chosen geometry:

- A hollow circular section, defined by an outer radius R and a thickness e.
- A rectangular section, defined by a height b and a width a.

In this study, the parameters R and e are varied on a logarithmic scale for a given material. The objective is to identify the pairs (R, e) that satisfy all the constraints imposed by the design requirements (maximum deflection, maximum stress, maximum mass, etc.).

Example for the maximum stress σ_{max}:

- The surface $\sigma_{max}(R, e)$ is plotted from the mechanical model.
- The validity zone is defined by $\sigma_{max} < \sigma_e$ (maximum allowable stress of the material).
- The same process is repeated for the other design criteria (deflection, mass...).

Each constraint defines an admissible region in the (R, e) plane. The intersection of all these regions gives a valid design domain. This synthesis is represented in a 2D chart (see figure below).

A pair (R, e) is then selected within this zone, according to a compromise between mechanical constraints and performance objectives (minimal mass, stiffness, robustness, etc.).

The following Python scripts were used to determine the optimal pairs:

```python
import numpy as np
import matplotlib.pyplot as plt
from matplotlib.patches import Patch
from matplotlib.lines import Line2D

# === CONSTANTES ===
L = 11.0            # Longueur de l'aile (m)
g = 9.81            # Gravit
M_tot = 500         # Masse totale en vol (kg)
v_max_seuil = 0.3 # Fl che max (m)
masse_max = 150     # Masse max du longeron (kg)

# Propri t s aluminium 7075-T6
E = 71.7e9          # Module Young [Pa]
sigma_e = 503e6     # Contrainte admissible [Pa]
rho = 2810          # Masse volumique [kg/m3]

# === FONCTIONS DE CALCUL ===
def calculer_contraintes_fleche_masse_rect(b, h):
    S = b * h
    I = (b * h**3) / 12
    M_a = rho * S * L

    q_sol = -rho * g * S
    p = M_tot * g / (2 * L)
    q_vol = p - rho * g * S
```

```python

    sigma_sol = abs((q_sol * L**2 / (2 * I)) * (h / 2))
    sigma_vol = abs((q_vol * L**2 / (2 * I)) * (h / 2))
    sigma_max = max(sigma_sol, sigma_vol)

    fleche_sol = abs((q_sol * L**4) / (8 * E * I))
    fleche_vol = abs((q_vol * L**4) / (8 * E * I))
    fleche_max = max(fleche_sol, fleche_vol)

    contrainte_ok = sigma_max < sigma_e
    fleche_ok = fleche_max < v_max_seuil
    masse_ok = M_a < masse_max

    return sigma_max, fleche_max, M_a, contrainte_ok, fleche_ok, masse_ok

# === GRILLE DE PARAM TRES ===
b_list = np.linspace(0.005, 0.05, 200)  # largeur b (m)
h_list = np.linspace(0.01, 0.3, 200)    # hauteur h (m)

b_grid, h_grid = np.meshgrid(b_list, h_list)

# === MATRICES DE CONTRAINTES ===
contrainte_ok = np.zeros_like(b_grid, dtype=bool)
fleche_ok = np.zeros_like(b_grid, dtype=bool)
masse_ok = np.zeros_like(b_grid, dtype=bool)

sigma_vals = np.zeros_like(b_grid)
fleche_vals = np.zeros_like(b_grid)
masse_vals = np.zeros_like(b_grid)

for i in range(len(h_list)):
    for j in range(len(b_list)):
        b = b_grid[i, j]
        h = h_grid[i, j]
        sigma, fleche, masse, c_ok, f_ok, m_ok = calculer_contraintes_fleche_masse_rect(b, h)

        contrainte_ok[i, j] = c_ok
        fleche_ok[i, j] = f_ok
        masse_ok[i, j] = m_ok

        sigma_vals[i, j] = sigma
        fleche_vals[i, j] = fleche
        masse_vals[i, j] = masse

# === ZONES DE DIMENSIONNEMENT ===
zones = np.zeros_like(b_grid, dtype=int)
zones[masse_ok] = 1
zones[masse_ok & contrainte_ok & ~fleche_ok] = 2
zones[masse_ok & fleche_ok & ~contrainte_ok] = 3
zones[masse_ok & contrainte_ok & fleche_ok] = 4

# === AFFICHAGE ===
plt.figure(figsize=(10, 8))
colors = ['black', 'red', 'yellow', 'orange', 'green']
labels = [
    'Aucune contrainte respect e',
```

```
    'Seulement masse OK',
    'Masse + contrainte m canique',
    'Masse + fl che OK',
    'Toutes contraintes OK'
]

zones_masked = np.ma.masked_where(b_grid >= h_grid, zones)

im = plt.contourf(b_grid, h_grid, zones, levels=np.arange(-0.5, 5.5, 1),
                  colors=colors, extend='neither')

plt.xlabel('Largeur a (m)', fontsize=12)
plt.ylabel('Hauteur b (m)', fontsize=12)
plt.title('Dimensionnement du longeron rectangulaire - Aluminium 7075-T6',
          fontsize=14, fontweight='bold')
plt.grid(True, alpha=0.3)



# === L GENDE ===
legend_elements = [Patch(facecolor=colors[i], label=labels[i]) for i in range(len(colors))]
legend_elements.append(Line2D([0], [0], marker='o', color='blue', label='Point optimal', markersize=8, linestyle='None'))

plt.legend(handles=legend_elements, loc='upper right', fontsize=10, frameon=True,
           fancybox=True, shadow=True, title="Zones et point optimal", title_fontsize=11)

plt.tight_layout()
plt.show()

# === AFFICHAGE DES R SULTATS POUR LE COUPLE CHOISI ===
a_choisi = 0.02  # largeur (m)
b_choisi = 0.20  # hauteur (m)

sigma_c, fleche_c, masse_c, c_ok, f_ok, m_ok = calculer_contraintes_fleche_masse_rect(a_choisi, b_choisi)

check = lambda cond: "   " if cond else "   "

print(f"\nPour le couple (a = {a_choisi:.2f} m, b = {b_choisi:.3f} m) :")
print(f" - Masse de l'aile        : {masse_c:.2f} kg {check(m_ok)}")
print(f" - Contrainte maximale    : {sigma_c/1e6:.2f} MPa {check(c_ok)}")
print(f" - Fl che maximale         : {fleche_c*100:.2f} cm {check(f_ok)}")
```

Listing 2: Python code for rectangular spar dimensioning

```
import numpy as np
import matplotlib.pyplot as plt
from math import pi

# Cahier des charges
L = 11.0   # longueur aile (m)
g = 9.81
```

```
M_tot = 500 # masse du planeur complet inf rieur    500 kg
M_n = 200  # masse nacelle + pilote (kg)
v_max_seuil = 0.3
masse_max = 150 # masse d'une aile qu'on ne doit pas d passer (kg)

# Propri t s Aluminium 7075-T6
E_alu = 71.7e9
rho_alu = 2810
sigma_e_alu = 505e6

def calculer_contraintes_fleche_masse(R, e):
    R_int = R - e
    if R_int <= 0:
        return float('inf'), float('inf'), float('inf'), False, False, False

    S = pi * (R**2 - R_int**2)
    I = (pi / 4) * (R**4 - R_int**4)
    M_a = rho_alu * S * L

    q_sol = -rho_alu * S * g
    p = M_tot * g / (2 * L)
    q_vol = p - rho_alu * S * g

    sigma_sol = abs((q_sol * L**2 / (2 * I)) * R)
    sigma_vol = abs((q_vol * L**2 / (2 * I)) * R)
    sigma_max = max(sigma_sol, sigma_vol)

    fleche_sol = abs((q_sol * L**4) / (8 * E_alu * I))
    fleche_vol = abs((q_vol * L**4) / (8 * E_alu * I))
    fleche_max = max(fleche_sol, fleche_vol)

    contrainte_ok = sigma_max < sigma_e_alu
    fleche_ok = fleche_max < v_max_seuil
    masse_ok = M_a < masse_max

    return sigma_max, fleche_max, M_a, contrainte_ok, fleche_ok, masse_ok

# Grille de param tres
e_list = np.logspace(-4, -1, 200)
R_list = np.logspace(-2, 0, 200)
e_grid, R_grid = np.meshgrid(e_list, R_list)

contrainte_ok = np.zeros_like(e_grid, dtype=bool)
fleche_ok = np.zeros_like(e_grid, dtype=bool)
masse_ok = np.zeros_like(e_grid, dtype=bool)

for i in range(len(R_list)):
    for j in range(len(e_list)):
        R = R_grid[i, j]
        e = e_grid[i, j]
        if e >= R:
            continue
        sigma_max, fleche_max, M_a, c_ok, f_ok, m_ok = calculer_contraintes_fleche_masse(R, e)
        contrainte_ok[i, j] = c_ok
        fleche_ok[i, j] = f_ok
```

```
        masse_ok[i, j] = m_ok

zones = np.zeros_like(e_grid, dtype=int)
zones[masse_ok] = 1
zones[masse_ok & contrainte_ok & ~fleche_ok] = 2
zones[masse_ok & fleche_ok & ~contrainte_ok] = 3
zones[masse_ok & contrainte_ok & fleche_ok] = 4

zones_masked = np.ma.masked_where(e_grid >= R_grid, zones)

plt.figure(figsize=(10, 8))

colors = ['black', 'red', 'yellow', 'orange', 'green']
labels = [
    'Aucune contrainte respect e',
    'Seulement masse OK',
    'Masse + contrainte m canique',
    'Masse + fl che OK',
    'Toutes contraintes OK'
]

im = plt.contourf(e_grid, R_grid, zones_masked, levels=np.arange(-0.5, 5.5, 1),
                  colors=colors, extend='neither')

plt.xlabel(' paisseur  e (m)', fontsize=12)
plt.ylabel('Rayon R (m)', fontsize=12)
plt.title('Dimensionnement du longeron - Aluminium 7075-T6\nZones autoris es (e,R)',
          fontsize=14, fontweight='bold')
plt.xscale('log')
plt.yscale('log')
plt.grid(True, alpha=0.3)

# Ligne limite physique e = R
e_line = np.logspace(-4, -1, 100)
plt.plot(e_line, e_line, 'k--', linewidth=2, alpha=0.8, label='Limite physique (e = R)')

# Ajout de lignes d'aide
plt.axvline(x=0.005, color='blue', linestyle='--', linewidth=1.5, label='e = 5 mm')
plt.axhline(y=0.10, color='blue', linestyle='--', linewidth=1.5, label='R = 10 cm')

from matplotlib.patches import Patch
legend_elements = [Patch(facecolor=colors[i], label=labels[i]) for i in range(len(colors))]
zone_legend = plt.legend(handles=legend_elements, loc='upper left', fontsize=10, frameon=True,
                         fancybox=True, shadow=True, title="Zones de dimensionnement", title_fontsize=11)

lines_legend = plt.legend(loc='lower right', fontsize=9, frameon=True,
                          fancybox=True, shadow=True, title="Rep res", title_fontsize=10)

plt.gca().add_artist(zone_legend)
```

```python
plt.tight_layout()
plt.show()

# === ANALYSE POUR LE COUPLE (R = 10 cm, e = 5 mm) ===
R_choisi = 0.10  # 10 cm
e_choisi = 0.005  # 5 mm

sigma_max, fleche_max, M_a, contrainte_ok, fleche_ok, masse_ok = calculer_contraintes_fleche_masse(R_choisi, e_choisi)

print("=== ANALYSE POUR L'ALUMINIUM 7075-T6 ===")
print(f"\nPour le couple (R = {R_choisi:.2f} m, e = {e_choisi:.3f} m) :")
print(f"  - Masse de l'aile         : {M_a:.2f} kg {'   ' if masse_ok else '   '}")
print(f"  - Contrainte maximale     : {sigma_max/1e6:.2f} MPa {'   ' if contrainte_ok else '   '}")
print(f"  - Fl che maximale          : {fleche_max*100:.2f} cm {'   ' if fleche_ok else '   '}")
```

Listing 3: Python code for circular spar dimensioning

From these results, we identify an admissible zone where the selected pairs meet all the design criteria:

$$(R, e) = (0.10, 0.005) \quad \text{and} \quad (a, b) = (0.02, 0.20)$$

2 Results for Two Geometric Configurations of the Spar

Table 7: Comparison of performances between two spar geometries

Parameters	**Hollow Circular Section**	**Rectangular Section**
Dimensions	$R = 0.10\,m$, $e = 0.005\,m$	$a = 0.02\,m$, $b = 0.20\,m$
Wing Mass	94.68 kg	123.64 kg
Maximum Stress	57.52 MPa	51.13 MPa
Maximum Deflection	24.27 cm	21.57 cm

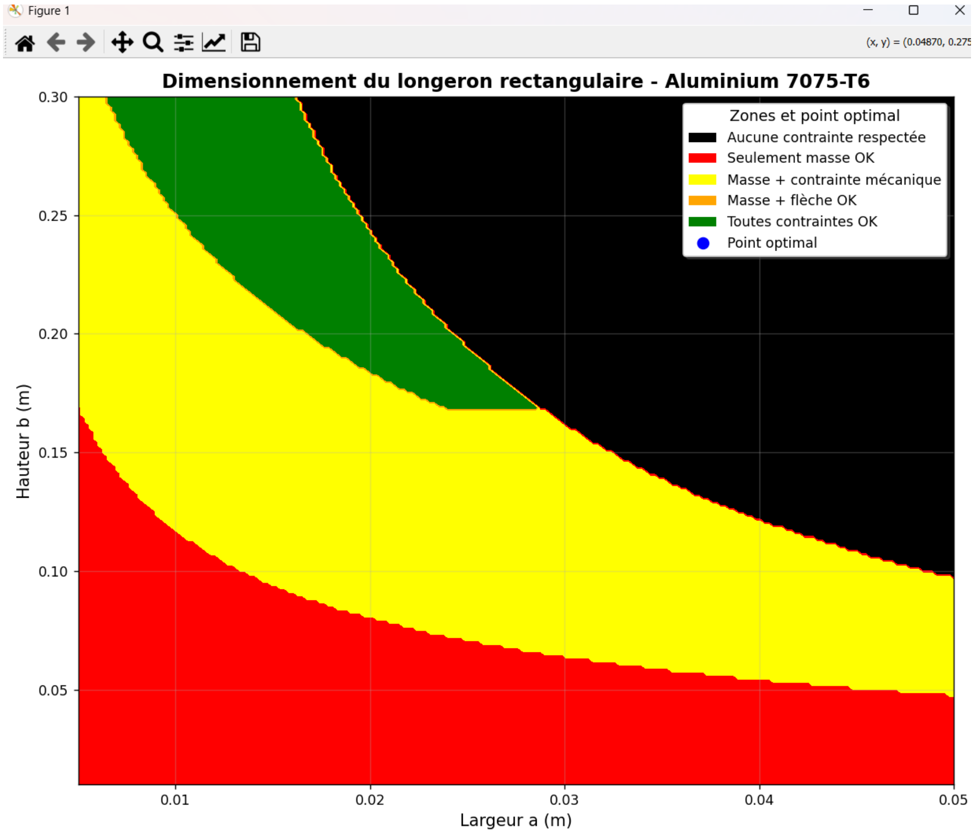


Figure V.1: Design zones for a rectangular spar section

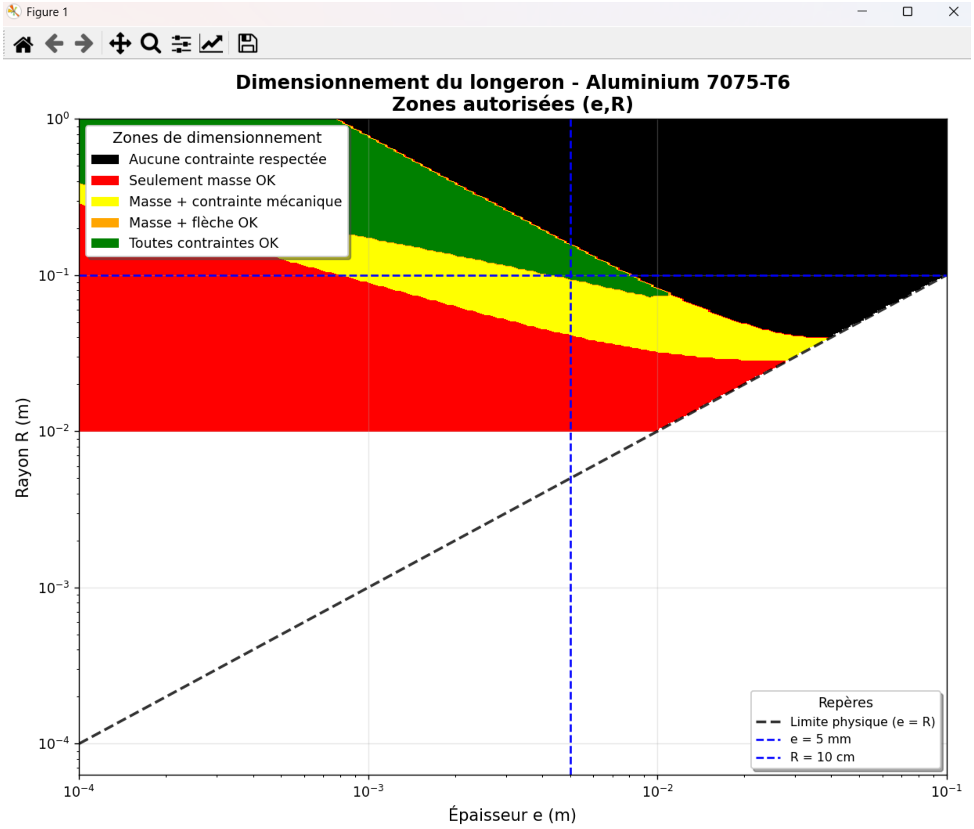


Figure V.2: Design zones for a rectangular spar section

VI Verification in Abaqus: Beam Model

We verify whether our analytical calculations are consistent with the beam model used in the finite element simulations.

1 Spar with Rectangular Section

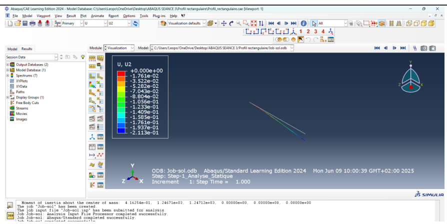
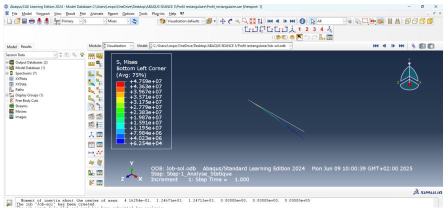
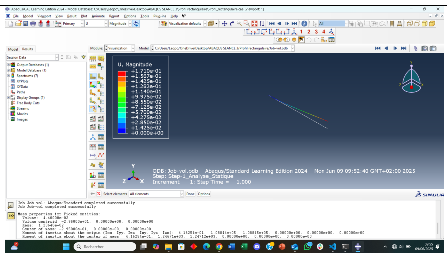
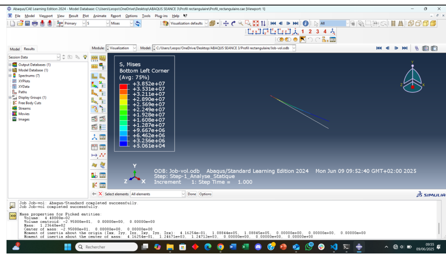
Simulation on Ground – Displacement	**Simulation on Ground – Von Mises Stress**
	
Maximum Deflection: $v_{\max} = 21.11\,\text{cm}$	**Maximum Stress:** $\sigma_{\max} = 47.6\,\text{MPa}$
Simulation in Flight – Displacement	**Simulation in Flight – Von Mises Stress**
	
Maximum Deflection: $v_{\max} = 17.1\,\text{cm}$	**Maximum Stress:** $\sigma_{\max} = 38.5\,\text{MPa}$

Table 8: Abaqus simulations for the rectangular spar: displacement and maximum Von Mises stress.

2 Spar with Circular Section

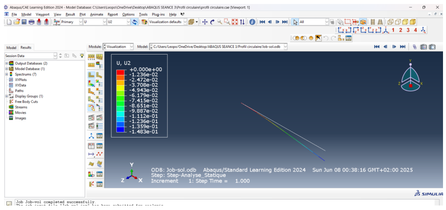
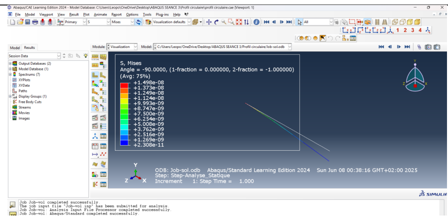
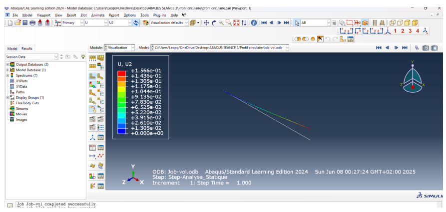
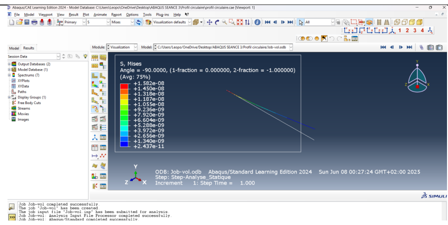
Simulation on Ground – Displacement	Simulation on Ground – Von Mises Stress
	
Maximum Deflection: $v_{\max} = 14.8\,\text{cm}$	**Maximum Stress:** $\sigma_{\max} = 230\,\text{MPa}$
Simulation in Flight – Displacement	**Simulation in Flight – Von Mises Stress**
	
Maximum Deflection: $v_{\max} = 15.1\,\text{cm}$	**Maximum Stress:** $\sigma_{\max} = 244\,\text{GPa}$

Table 9: Abaqus simulations for the circular spar: displacement and maximum Von Mises stress.

3 Summary Table

Model	Condition	Quantity	Theoretical	Abaqus	Difference
Rectangular Section	Ground	**Max Deflection** $v_{\max}$	**21.1 cm**	**21.11 cm**	–
		Max Stress $\sigma_{\max}$	**50 MPa**	**47.6 MPa**	–
		Mass	**123.64 kg**	**123.64 kg**	–
	Flight	**Max Deflection** $v_{\max}$	**21.6 mm**	**17.1 cm**	–
		Max Stress $\sigma_{\max}$	**51.1 MPa**	**38.5 MPa**	–
		Mass	**123.64 kg**	**123.64 kg**	–
Circular Section	Ground	**Max Deflection** $v_{\max}$	**14.79 cm**	**14.8 cm**	–
		Max Stress $\sigma_{\max}$	**35 MPa**	**230 MPa**	–
		Mass	**94.68 kg**	**94.68 kg**	–
	Flight	**Max Deflection** $v_{\max}$	**24.27 cm**	**15.1 cm**	–
		Max Stress $\sigma_{\max}$	**57.52 MPa**	**244 GPa**	–
		Mass	**94.68 kg**	**94.68 kg**	–

Table 10: Comparison between theoretical and Abaqus results for both spar configurations.

4 Analysis of Results

By comparing the Abaqus simulations with theoretical predictions, the results can be evaluated against the design requirements:

- The maximum allowed deflection is 50 cm ($v_{\max} \leqslant 0.5$ m);
- The stress must remain below the elastic limit of aluminium 7075 (250 MPa);
- The spar mass should remain as low as possible.

Rectangular spar: All constraints are met.

- Deflections: 21.11 cm (ground) and 17.1 cm (flight) $\Rightarrow$ both below 50 cm.
- Maximum stresses: 47.6 MPa (ground) and 38.5 MPa (flight) $\Rightarrow$ below the elastic limit.
- Mass: 123.64 kg $\Rightarrow$ within target range.

Circular spar: Shows inconsistent results.

- Deflections: 14.8 cm (ground) and 15.1 cm (flight) $\Rightarrow$ acceptable.
- Stresses: 230 MPa (ground) and 244 GPa (flight) $\Rightarrow$ the latter is unrealistic (simulation/unit error).
- Mass: 94.68 kg $\Rightarrow$ lighter but with unreliable stress results.

Conclusion: The rectangular spar fully satisfies all design criteria (stiffness, strength, mass). The circular spar, though lighter, exhibits unrealistic stress under flight conditions and is unsuitable in its current configuration.

VII Conclusion

This report detailed the different stages of the mechanical sizing of a glider wing, focusing on the choice of thickness and material, as well as the sizing of the spar. The main objectives were to analyze geometric and mechanical constraints, determine the optimal wing thickness, evaluate mechanical performance, select the appropriate material, and dimension the spar.

Both theoretical models and numerical simulations must be used complementarily to ensure safety, performance, and efficiency. The results show that **Aluminium 7075-T6** is a suitable material for wing design, and that simplified analytical models provide valuable insights, although advanced simulations are required for full validation